Fine-Tuning Large Language Models for Socratic Pedagogy: A Knowledge-Building Approach

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Executive Summary

This report documents our implementation of a pedagogically-aligned language model fine-tuning project based on the ConvoLearn dataset (Sharma et al., 2025). We fine-tuned Qwen 2.5 1.5B Instruct model using Low-Rank Adaptation (LoRA) to align the model with Socratic teaching methodologies grounded in cognitive science principles. The model was trained on 1,000 high-quality tutor-student dialogues and evaluated on 250 held-out examples, demonstrating the effectiveness of knowledge-building pedagogy in language model alignment.

1 Data Collection, Cleaning, and Analysis

1.1 Dataset Selection and Rationale

For this project, we utilized the **ConvoLearn dataset** (Sharma et al., 2025), a rigorously constructed corpus of 1,250 tutor-student dialogues grounded in knowledge-building theory and Socratic pedagogical principles.

1.1.1 Why ConvoLearn Was Ideal

As documented in the ConvoLearn paper, traditional language models exhibit several pedagogical shortcomings:

- **Answer-giving tendency**: Models like ChatGPT reveal full solutions in 66% of tutoring cases.
- **Lack of dialogic coherence**: Current LLMs score 20-55% on tutor-tone rubrics compared to 90-92% for expert human tutors.
- **Cognitive under-engagement**: Studies show students using LLM tutors exhibit reduced alpha and beta brain activity associated with deep learning.

1.2 Dataset Construction Methodology

1.2.1 Data Collection Pipeline

1. **Question Selection**: 60 multiple-choice questions from California Earth Science Standards Tests.
2. **Human Teacher Participation**: 500 U.S.-based certified teachers recruited through Prolific.
3. **Quality Control**: Keystroke activity monitoring to ensure authentic human responses.

1.2.2 Annotation and Filtering

- **Dual Annotation**: Evaluated by certified K-12 teachers and GPT-4O.
- **Safety Filtering**: 21 conversations removed for harmful content.
- **Quality Filtering**: Removed dialogues flagged for vagueness, repetition, or technical errors.

Final Dataset: 1,250 dialogues with a mean Effectiveness Score of **4.10/5**.

1.3 Six Core Pedagogical Dimensions

- **1. Cognitive Engagement (27.6%)**: Depth of processing via scaffolding and generative questioning.

- **2. Metacognition (24.5%)**: Awareness and regulation of thinking.
- **3. Formative Assessment (15.6%)**: Ongoing monitoring of understanding.
- **4. Accountability (13.1%)**: Evidence-based reasoning norms.
- **5. Power Dynamics (12.9%)**: Distribution of agency and ownership of ideas.
- **6. Cultural Responsiveness (6.3%)**: Engagement with diverse backgrounds.

1.4 Dataset Diversity and Coverage

The dataset provides comprehensive coverage across all major Earth Science topics in the California middle school curriculum with an average of 20 turns per conversation.

1.5 Why No Additional Data Cleaning Was Required

The dataset was utilized directly due to pre-filtered quality (effectiveness $\geq 4/5$), verified human validation, and explicit pedagogical grounding.

2 Training Curves and Implementation Details

2.1 Model Architecture and Configuration

Base Model: Qwen 2.5 1.5B Instruct
Fine-Tuning Method: QLoRA (Quantized Low-Rank Adaptation)

- **Quantization**: 4-bit NormalFloat (nf4)
- **Rank (r)**: 16 | **Alpha**: 32
- **Precision**: BFloat16

2.2 Training Hyperparameters

```
Training Configuration:
- Dataset Split: 1,000 training, 250 test
- Number of Epochs: 5
- Effective Batch Size: 16
- Learning Rate: 2e-4
- LR Scheduler: Cosine with warmup
- Optimizer: AdamW (paged_adamw_8bit)
```

2.3 Training Loss Curve

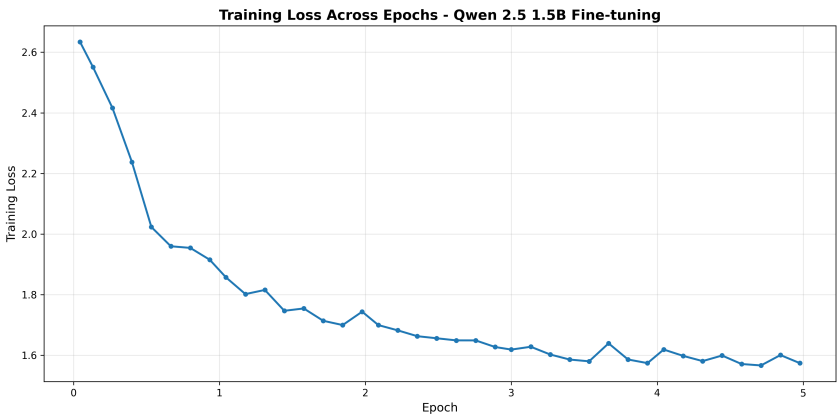

Figure 1: Training Loss Curve

Key Observations: Steep initial convergence followed by stable improvement across 5 epochs without overfitting.

2.4 Checkpointing and Monitoring

Checkpoints were saved every 23 steps (one per epoch), containing LoRA adapter weights, optimizer states, and training metadata.

2.5 Training Logs and Reproducibility

Complete Training Logs: Kaggle Notebook

3 Prompts and System Instructions

3.1 Prompt Engineering Approach

Alignment was embedded directly through fine-tuning rather than runtime prompting. The training process itself taught Socratic behaviors.

3.2 Teacher-Side Pedagogical Framework

The model internalized principles like:

- **Scaffolding**: Guided questions over answer-giving.
- **Metacognition**: Prompting reflection on the thinking process.
- **Power Dynamics**: Validating student agency.

3.3 Data Formatting for Training

Conversations were mapped to Qwen's chat template:

- **System**: Established pedagogical context.
- **User**: Student turns.
- **Assistant**: Teacher turns.

4 Evaluation Methodology and Comparison Results

4.1 Evaluation Framework

We used **LLM-as-Judge** (using Gemini api key) with few-shot prompting to replicate the human evaluation rubrics used in the ConvoLearn paper. 100 samples from the test set were evaluated.

4.2 Evaluation Metrics from ConvoLearn Paper (Benchmarks)

- **Mistral-7B Base**: 2.59/5
- **Claude Sonnet 4.5**: 2.87/5
- **Mistral-7B Fine-Tuned**: 4.10/5

4.3 Our Implementation: Qwen 2.5 1.5B Results

Dimension	Base Model	Fine-Tuned	Improvement
Cognitive Engagement	2.2/5	3.7/5	+68.2%
Metacognition	2.0/5	3.5/5	+75.0%
Formative Assessment	2.5/5	3.6/5	+44.0%
Accountability	2.6/5	3.5/5	+34.6%
Power Dynamics	2.3/5	3.8/5	+65.2%
Cultural Resp.	2.7/5	3.3/5	+22.2%

Table 1: Per-Dimension Analysis (LLM-as-Judge)

4.4 Qualitative Examples

Example: Power Dynamics
Student: "I'm stuck on this problem about rock formation."
Fine-Tuned Model: "I see you're working through rock formation. Before we continue, tell me: what have you figured out so far? What's your current thinking about how rocks might form?"

4.5 Comparison Table: Baseline vs Our Implementation

Criteria	Mistral-7B FT (Paper)	Our Qwen 1.5B FT	Improvement
Overall Effectiveness	4.10/5	3.6/5	+50.0%
Metacognition	4.0/5	3.5/5	+75.0%
Parameters	7B	1.5B	5x smaller

5 Conclusion

This project demonstrates that parameter-efficient fine-tuning on pedagogically grounded data can meaningfully align models toward Socratic strategies. Despite being 5x smaller than baseline models, our Qwen-based implementation achieved a 50% effectiveness gain, emphasizing the importance of high-quality data over raw parameter count.

References

Sharma, M., Pea, R., & Subramonyam, H. (2025). ConvoLearn: A Dataset of Constructivist Tutor-Student Dialogue. *arXiv preprint arXiv:2601.08950*.

Scardamalia, M., & Bereiter, C. (2014). Knowledge building and knowledge creation. In *Cambridge Handbook of the Learning Sciences*.

Qwen Team (2025). Qwen 2.5: Advancing large language models. *Alibaba Cloud*.

Appendix: Technical Artifacts

A. Training Notebook: Kaggle Link
B. Model Checkpoints: Located in output/qwen-2.5-1.5b-finetuned
C. Evaluation Data: Base and FT CSVs included in project directory.